

4K/UHD 8x4 HDBaseT™ and HDMI Matrix Switcher

Application Programming Interface
0.1.40

Version Information

Version	Release Date	Notes
1	09/18	Initial release

Commands

The following tables provide an alphabetical list of commands available on the AT-UHD-CLSO-840. All commands are case-sensitive and must be entered as documented. If the command fails or is entered incorrectly, then the feedback is “Command FAILED”.



IMPORTANT: Each command is terminated with a carriage-return (0x0d) and the feedback is terminated with a carriage-return and line-feed (0x0a).

Command	Description
All#	Resets the matrix to a one-to-one routing state
AudioSrc	Sets the audio output channel for analog or digital audio
Blink	Enables or disables blinking of the POWER button on the front panel
Broadcast	Enables or disables broadcast mode
Clear	Erases the specified preset
CSpara	Sets the baud rate, data bits, parity bit, and stop bits for the serial port
EDIDMSet	Assigns the specified EDID to the desired input
EDIDOut	Saves the downstream EDID to the specified memory location
HDCPSet	Sets the HDCP reporting mode for the specified input
help	Displays the available list of commands
HTTPPort	Sets the HTTP listening port
InputBroadcast	Enables or disables broadcast mode
InputStatus	Displays the status for each input
IPAddUser	Adds a user for Telnet control
IPCFG	Displays the current network settings for the AT-UHD-CLSO-840
IPChangePass	Changes the password for the web GUI and Telnet sessions
IPDelUser	Deletes the specified Telnet user
IPDHCP	Enables or disables DHCP mode on the AT-UHD-CLSO-840
IPLogin	Enables or disables login credentials when starting a Telnet session
IPPort	Sets the Telnet listening port for the AT-UHD-CLSO-840
IPQuit	Terminates the current TCP/IP session
IPStatic	Sets the static IP address, subnet mask, and gateway for the AT-UHD-CLSO-840
IPTimeout	Specifies the time interval of inactivity before the Telnet session is closed
Lock	Locks the buttons on the front panel
LockST	Displays the lock status of the AT-UHD-CLSO-840
MirrorAudio	Enables / disables mirroring for the specified audio and video outputs
Mreset	Resets the AT-UHD-CLSO-840 to factory-default settings
PWOFF	Places the AT-UHD-CLSO-840 in standby mode
PWON	Powers-on the matrix
PWSTA	Displays the current power state

Commands

Command	Description
Recall	Recalls the specified routing preset
RS232para	Sets the baud rate, data bits, stop bits, and parity for the RS-232 port
RS232zone	Send a command to the RS-232 device connected to the RS-232 port on the receiver
Save	Saves the current routing state to the specified preset
Status	Displays which inputs are connected to which outputs
TrigCEC	Triggers the stored CEC command
Type	Displays the model of the transmitter
Unlock	Unlocks the buttons on the front panel
Version	Displays the current firmware version of the AT-UHD-CLSO-840
VIN	Specifies the volume input level
VOUT	Specifies the volume input level
VOUTMute	Mutes the audio on the specified channel (zone)
x?\$	Enables / disables the specified output channel
x?All	Routes the specified input to all outputs
x?AVx?	Routes an input to the specified output(s)
x?Ax?	Routes the audio from a video input to the specified audio output

All#

Resets the matrix to the factory-default one-to-one routing state. This command only resets the input/output routing: Input 1 > Output 1, Input 2 > Output 2, etc. No other settings are affected.

Syntax

All#

This command does not require any parameters

Example

All#

Feedback

All#

AudioSrc

Specifies the audio type to be output on the output channel (HDMI 1, HDMI 2, HDBaseT 3, HDBaseT 4).

Syntax

AudioSrcX Y

Parameter	Description	Range
X	Output Channel	1 ... 4
Y	Audio Type	ana, dig

Example

AudioSrc2 ana

Feedback

AudioSrc2 ana

Blink

Enables or disables blinking of the **POWER** button on the front panel. When set to on, the **POWER** button will flash, alternating between red and blue, and can be used to physically identify the unit on a network. The **POWER** button will flash until the Blink off command is executed. on = enables blinking; off = disables blinking; sta = displays the current setting. The default setting is off.

Syntax

Blink X

Parameter	Description	Range
X	Value	on, off, sta

Example

Blink on

Feedback

Blink on

Broadcast

Enables or disables broadcast mode. By default, broadcast mode is set to off. When set to on, changes in the web GUI will also be affected on the control system (if connected), via TCP/IP. To separate control between web GUI and Telnet, set this feature off. on = enables broadcast mode; off = disables broadcast mode; sta = displays the current setting.

Syntax

Broadcast X

Parameter	Description	Range
X	Value	on, off, sta

Example

Broadcast on

Feedback

Broadcast on

Clear

Erases the specified preset.

Syntax

ClearX

Parameter	Description	Range
X	Preset	1 ... 5

Example

Clear2

Feedback

Clear2

CSpa

Sets the baud rate, data bits, parity bit, and stop bits for the serial port. Use the sta argument to display the current serial port settings. Each argument must be separated by a comma; no spaces are permitted. Brackets must be used when executing this command.

Syntax

CSpa[W,X,Y,Z]

Parameter	Description	Range
W	Baud rate	2400, 4800, 9600, 19200, 38400, 57600, 115200
X	Data bits	7, 8
Y	Parity bit	None, Odd, Even
Z	Stop bits	1, 2

Example

CSpa[115200,8,0,1]
CSpa[sta]

Feedback

CSpa[115200,8,0,1]
CSpa [115200,8,0,1]

EDIDMSet

Assigns an EDID to the specified input. The EDID can be one of the internal pre-programmed EDID selections or a custom EDID that can be stored in one of the four memory locations. A brief description of each pre-programmed EDID is listed in the table below. For a detailed summary of each EDID, refer to the Appendix in the User Manual table. Use arguments save1 through save4 to store the EDID in any of four memory locations. To display the EDID assigned to an input, use the sta argument. Do not use a space between the command and the first parameter.

For the first parameter, inputs are represented as follows:

1 = HDMI IN 1
2 = HDMI IN 2
3 = VGA IN

4 = DP IN 4
5 = HDMI IN 5 (receiver)

Syntax

EDIDMSetX Y

Parameter	Description	Range
X	Input	1 ... 5
Y	EDID	default, save1 ... save4, int1 ... int23, sta

Example

EDIDMSet3 int6
EDIDMSet3 sta

Feedback

EDIDMSet3 int6
EDIDMSet3 int6

EDIDOut

Save the EDID of the display (sink), connected to the receiver, to internal memory. The first parameter must be 1.

Syntax

```
EDIDOutX Y
```

Parameter	Description	Range
X	Output	1 (required)
Y	Memory location	save1 ... save4

Example

```
EDIDOut1 save2
```

Feedback

```
EDIDOut1 save2
```

HDCPSet

Set the HDCP reporting mode of the specified port. Some computers will send HDCP content if an HDCP-compliant display is detected. on = reports to the source device that the display (sink) is HDCP-compliant, off = reports to the source device that the display (sink) is not HDCP-compliant (HDCP content will not be sent), auto = uses the attributes of the display device to accept or not accept HDCP content. Setting this value to off *does not* decrypt HDCP content. Use the sta argument to display the current setting.

Syntax

```
HDCPSet X
```

Parameter	Description	Range
X	Value	on, off, auto, sta

Example

```
HDCPSet 1 on
```

Feedback

```
HDCPSet 1 on
```

help

Displays the list of available commands. To obtain help on a specific command, enter the **help** command followed by the name of the command.

Syntax

```
help [X]
```

Parameter	Description	Range
X	Command name (optional)	Command

Example

```
help
```

Feedback

Available cmds:

```
help
PWON
PWOFF
PWSTA
...
...
```

HTTPPort

Sets the HTTP listening port for the AT-UHD-CLSO-840.

Syntax

```
HTTPPort X
```

Parameter	Description	Range
X	Port	0 ... 65535

Example

```
HTTPPort 8080
```

Feedback

```
HTTPPort 8080
```

InputBroadcast

Enables or disables broadcast mode. The default setting is off.

Syntax

```
InputBroadcast X
```

Parameter	Description	Range
X	State	on, off, sta

Example

```
InputBroadcast on
```

Feedback

```
InputBroadcast on
```

InputStatus

Displays the status of each input as a set of five consecutive numbers between 0 and 1. Each number represent an input, from 1 to 5: **HDMI IN 1**, **HDMI IN 2**, **VGA IN**, **DP IN 4**, and **HDMI IN 5**, respectively. If a source is detected on the input, then a 1 will be displayed. Inputs with no source connected will display a 0.

Syntax

```
InputStatus
```

This command does not require any parameters

Example

```
InputStatus
```

Feedback

```
InputStatus 10010
```

IPAddUser

Adds a user for web GUI login and Telnet sessions. This command performs the same function as adding a user within the web GUI. Refer to User Manual for more information.

Syntax

```
IPAddUser X Y
```

Parameter	Description	Range
X	User name	20 characters (max)
Y	Password	20 characters (max)

Example

```
IPAddUser BigBoss b055man
```

Feedback

```
IPAddUser BigBoss b055man
TCP/IP user was added
```

IPCFG

Displays the current network settings for the AT-UHD-CLSO-840.

Syntax

```
IPCFG
```

This command does not require any parameters

Example

```
IPCFG
```

Feedback

```
MAC addr: B8:98:B0:03:1D:7C
IP addr: 10.0.1.182
Netmask: 255.255.255.0
Gateway: 10.0.1.1
Web Port: 80
Telnet Port: 23
```

```
Send with TCP/IP 1:
IP Address: 192.168.1.10
TCP Port: 23
Listen UDP Port: 8
Send with: TCP
```

```
Send with TCP/IP 2:
IP Address: 192.168.1.11
TCP Port: 23
Listen UDP Port: 18
Send with: TCP
```

IPChangePass

Changes the password for the specified user.

Syntax

```
IPChangePass X
```

Parameter	Description	Range
W	Username	--
X	Old password	--
Y	New password	--
Z	Password confirmation	--

Example

```
IPChangePass root Atlona BoS5man
```

Feedback

```
IPChangePass root Atlona BoS5man
```

IPDelUser

Deletes the specified user. Deleted users will no longer be able to access the web GUI or initiate Telnet sessions. This command performs the same function as removing a user within the web GUI. Refer to the User Manual for more information.

Syntax

```
IPDelUser X
```

Parameter	Description	Range
X	User	User name

Example

```
IPDelUser Minion2
```

Feedback

```
IPDelUser Minion2
TCP/IP user was deleted
```

IPDHCP

Enables or disables DHCP mode on the AT-UHD-CLSO-840. on = DHCP mode ON; off = DHCP mode OFF; sta = displays the current setting. If this feature is disabled, then a static IP address must be specified. The default setting is DHCP = ON.

Syntax

```
IPDHCP X
```

Parameter	Description	Range
X	Value	on, off, sta

Example

```
IPDHCP on
```

Feedback

```
IPDHCP on
```

IPLogin

Enables or disables the use of login credentials when initiating a Telnet session on the AT-UHD-CLSO-840. If this feature is set to on, then the AT-UHD-CLSO-840 will prompt for both the username and password. Use the same credentials as the web GUI. on = login credentials required; off = no login required. Use the sta argument to display the current setting. The default setting is on.

Syntax

```
IPLogin X
```

Parameter	Description	Range
X	Value	on, off, sta

Example

```
IPLogin off
```

Feedback

```
IPLogin off
```

IPPort

Sets the TCP/IP listening port for the AT-UHD-CLSO-840.

Syntax

```
IPPort X
```

Parameter	Description	Range
X	Port	0 ... 65535, sta

Example

```
IPPort 230
```

Feedback

```
IPPort 230
```

IPQuit

Closes the current Telnet session.

Syntax

```
IPQuit
```

This command does not require any parameters

Example

```
IPQuit
```

Feedback

```
Connection lost...
```

IPStatic

Sets the static IP address, subnet mask, and gateway (router) address of the AT-HDVS-210U-TX-WP. Before using this command, DHCP must be disabled on the AT-HDVS-210U-TX-WP. Refer to the [IPDHCP](#) command for more information. Each argument must be entered in dot-decimal notation and separated by a space. The default static IP address of the AT-UHD-CLSO-840 is 192.168.1.254.

Syntax

```
IPStatic X Y Z
```

Parameter	Description	Range
X	IP address	0 ... 255 (per byte)
Y	Subnet mask	0 ... 255 (per byte)
Z	Gateway (router)	0 ... 255 (per byte)

Example

```
IPStatic 192.168.1.112 255.255.255.0 192.168.1.1
```

Feedback

```
IPStatic 192.168.1.112 255.255.255.0 192.168.1.1
```

IPTimeout

Specifies the time interval of inactivity before the TCP/IP session is terminated. When terminated, both the Telnet and web GUI session will be closed. The default setting is 300 seconds.

Syntax

```
IPTimeout X
```

Parameter	Description	Range
X	Interval (in seconds)	1 ... 60000

Example

```
IPTimeout 300
```

Feedback

```
IPTimeout 300
```

Lock

Locks the buttons on the front panel. This feature is useful when the unit is installed in a rack environment or other remote location, to prevent unauthorized tampering or accidental pressing of the front-panel buttons. Also refer to the **Unlock** command.

Syntax

```
Lock
```

This command does not require any parameters

Example

```
Lock
```

Feedback

```
Lock
```

LockST

Displays the current lock status.

Syntax

```
LockST
```

This command does not require any parameters

Example

```
LockST
```

Feedback

```
Unlock
```

MirrorAudio

Enables or disables mirroring for the specified audio and video outputs. Refer to the User Manual for more information on mirroring. Specify 0 for the second parameter to disabling mirroring.

Syntax

```
MirrorAudioX Y
```

Parameter	Description	Range
X	Audio Output	1 ... 4
Y	Video Output	0, Out 1 ... Out 4

Example

```
MirrorAudio2 Out3
```

Feedback

```
MirrorAudio2 Out3
```

Mreset

Resets the AT-UHD-CLSO-840 to factory-default settings.

Syntax

```
MReset
```

This command does not require any parameters

Example

```
Mreset
```

Feedback

```
Mreset
```

PWOff

Executing this command will power-off the AT-JUNO-451. Use the PWON command to power-on the unit.

Syntax

```
PWOff
```

This command does not require any parameters

Example

```
PWOff
```

Feedback

```
PWOff
```

PWON

Executing this command will power-on the AT-JUNO-451. Use the PWOFF command to power-off the unit.

Syntax

PWON

This command does not require any parameters

Example

PWON

Feedback

PWON

PWSTA

Displays the current power state of the AT-JUNO-451.

Syntax

PWSTA

This command does not require any parameters

Example

PWSTA

Feedback

PWON

Recall

Loads the specified routing preset.

Syntax

Recall

Parameter	Description	Range
X	Preset	1 ... 5

Example

Recall2

Feedback

Recall2

RS232para

Sets the baud rate, data bits, parity bit, and stop bits for the **RS-232** port on the AT-UHD-CLSO-840. Each argument must be separated by a comma; no spaces are permitted. Brackets must be included when typing this command.

Syntax

```
RS232paraV[W,X,Y,Z]
```

Parameter	Description	Range
V	Zone	1 ... 5
W	Baud rate	2400, 9600, 19200, 38400, 56000, 57600, 115200
X	Data bits	8, 9
Y	Parity bit	0 (None), 1 (Odd), 2 (Even)
Z	Stop bits	1, 2

Example

```
RS232para1[115200,8,0,1]
```

Feedback

```
RS232para1[115200,8,0,1]
```

RS232zone

Sends commands to the connected display. Refer to the User Manual of the display device for a list of available commands. Brackets must be used when specifying the command argument. The command line must not contain any spaces.

Syntax

```
RS232zone[X][Y]
```

Parameter	Description	Range
X	Output port (zone)	1, 2
Y	Command	String

Example

```
RS232zone1[command]
```

Feedback

```
RS232zone1[command]
```

Save

Saves the current routing state to the specified preset.

Syntax

Save

Parameter	Description	Range
X	Preset	1 ... 5

Example

Save2

Feedback

Save2

Status

Displays the current routing status for each A/V and audio port.

Syntax

Status

This command does not require any parameters

Example

Status

Feedback

x1AVx1,x2AVx2,x3AVx3,x4AVx4,x1Ax1,x3Ax2,x3Ax3,x4Ax4

TrigCEC

Sends the specified command to the display using CEC. The output must always be specified and set to the value of 1. Do not add a space between the command and the first argument.

Syntax

TrigCECX Y

Parameter	Description	Range
X	Zone	1
Y	Power state	on, off

Example

TrigCEC1 on

Feedback

TrigCEC1 on

Type

Displays the model information of the AT-UHD-CLSO-840. The full argument may be specified to display the host name and firmware version. This argument is optional.

Syntax

Type X

Parameter	Description	Range
X	Information (optional)	full

Example
Type

Feedback
AT-UHD-CLSO-840

Unlock

Unlocks the buttons on the front panel. Also refer to the [Lock](#) command.

Syntax

Lock

This command does not require any parameters

Example
Lock

Feedback
Lock

Version

Displays the current version of firmware.

Syntax

Version

This command does not require any parameters

Example
Version

Feedback
Firmware 0.1.55

VIN

Sets the input volume, in decibels, for the specified input. Values for parameter Y can be specified numerically or can be incremental/decremental using the + or - character.

Syntax

VINX Y

Parameter	Description	Range
X	Input	1 ... 4
Y	Value	-10 ... 10, +, -

Example

VIN1 2

Feedback

VIN1 2

VOUT

Sets the output channel volume, in decibels. Values for parameter Y can be specified numerically or can be incremental/decremental using the + or - character.

Syntax

VOUTX Y

Parameter	Description	Range
X	Output	1 ... 4
Y	Value	-79 ... 15, +, -

Example

VOUT2 +

Feedback

VOUT2 -9

VOUtmute

Controls volume muting for the specified audio channel (zone). Use the sta argument to display the current muting state for the specified zone.

Syntax

```
VOUtmuteX Y
```

Parameter	Description	Range
X	Zone	1 ... 4
Y	State	off, on, sta

Example

```
VOUtmute2 on
```

Feedback

```
VOUtmute2 on
```

x?\$

Enables or disables the specified output channel. Use the sta argument to display the current state.

Syntax

```
xX$ Y
```

Parameter	Description	Range
X	Channel	1 ... 4
Y	State	off, on, sta

Example

```
x3$ off
```

Feedback

```
x3$ off
```

x?All

Routes the specified input to all outputs.

Syntax
xXAll

Parameter	Description	Range
X	Channel	1 ... 4

Example
x2All

Feedback
x2All

x?AVx?

Routes the specified input to the specified output. Multiple outputs may be specified. If multiple outputs are specified, use the comma delimiter with a spaces (as shown in the example below).

Syntax
xXAVxY

Parameter	Description	Range
X	Input	1 ... 8
Y	Output	1 ... 4

Example
x3AVx2, x4, x7

Feedback
x3AVx2, x4, x7

x?Ax?

Routes the audio from the specified video input to the specified audio output.

Syntax
xXAxY

Parameter	Description	Range
X	Video Input	1 ... 8
Y	Video Output	1 ... 4

Example
x6Ax2

Feedback
x6AVx2

